AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 1 · STUDENT LESSON

Predictive and Generative AI

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Name the target. Trace the stages. Check each claim.

Revisit prediction and generation in a source-grounded study workflow.

Keep one habit: Component contract → corpus distribution → draft audit → evidence plan for a new workflow.

MAIN LESSON · SECTION 1

Which component produced this output, and what checks it?

> Revisit prediction and generation through a source-grounded study tool.

Your learning goals

I can distinguish discriminative **prediction** from generation by inspecting the requested **output**.

I can calculate an empirical next-token distribution from counted continuations.

I can separate next-token fit from factual support and downstream usefulness.

Recall the capstone evidence boundary

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- What is the difference between a model **prediction** and **evidence** that a statement is true?

Your end goal

By the end, I can identify generation and **prediction** objectives without confusing either with verified truth.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

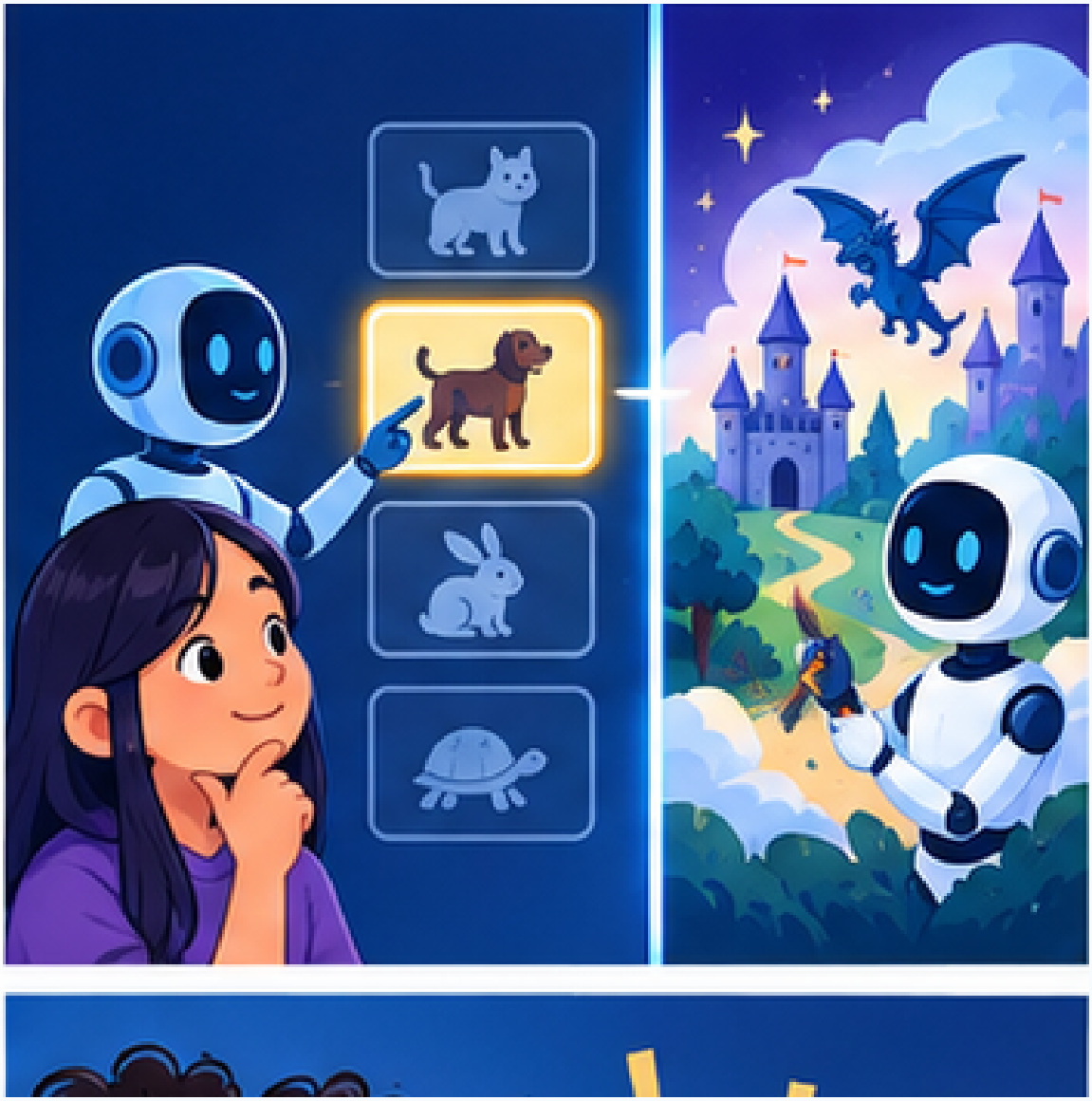
Remember before reading

Without opening the old lesson, explain one key idea from Capstone Expo. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 1: Predictive and **Generative AI**

Prediction estimates a **target**; generation produces structured **output**. Autoregressive generation predicts successive items and may choose them by sampling or a declared decoding rule.
</details>



CREATOR MISSION

Classify systems and explain the **output space** for each.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Describe the task and permitted output of each component

> Generation makes predictions too; whole products combine mechanisms.

Targets and output spaces

A classifier might map a message to the label request or report. A text generator instead produces a sequence, predicting a distribution over the next token given a prefix.

Both make **prediction**s, but the **training target**s and permitted **output**s differ. One application can combine generation with a classifier, retrieval and human approval.

Calling a whole product “generative” does not describe every component.

Next-token fit is not a factual certificate

The objective defines what **training** attempts to fit. Next-token fit rewards matching observed continuations; it does not directly certify that an answer is true, well sourced or permitted.

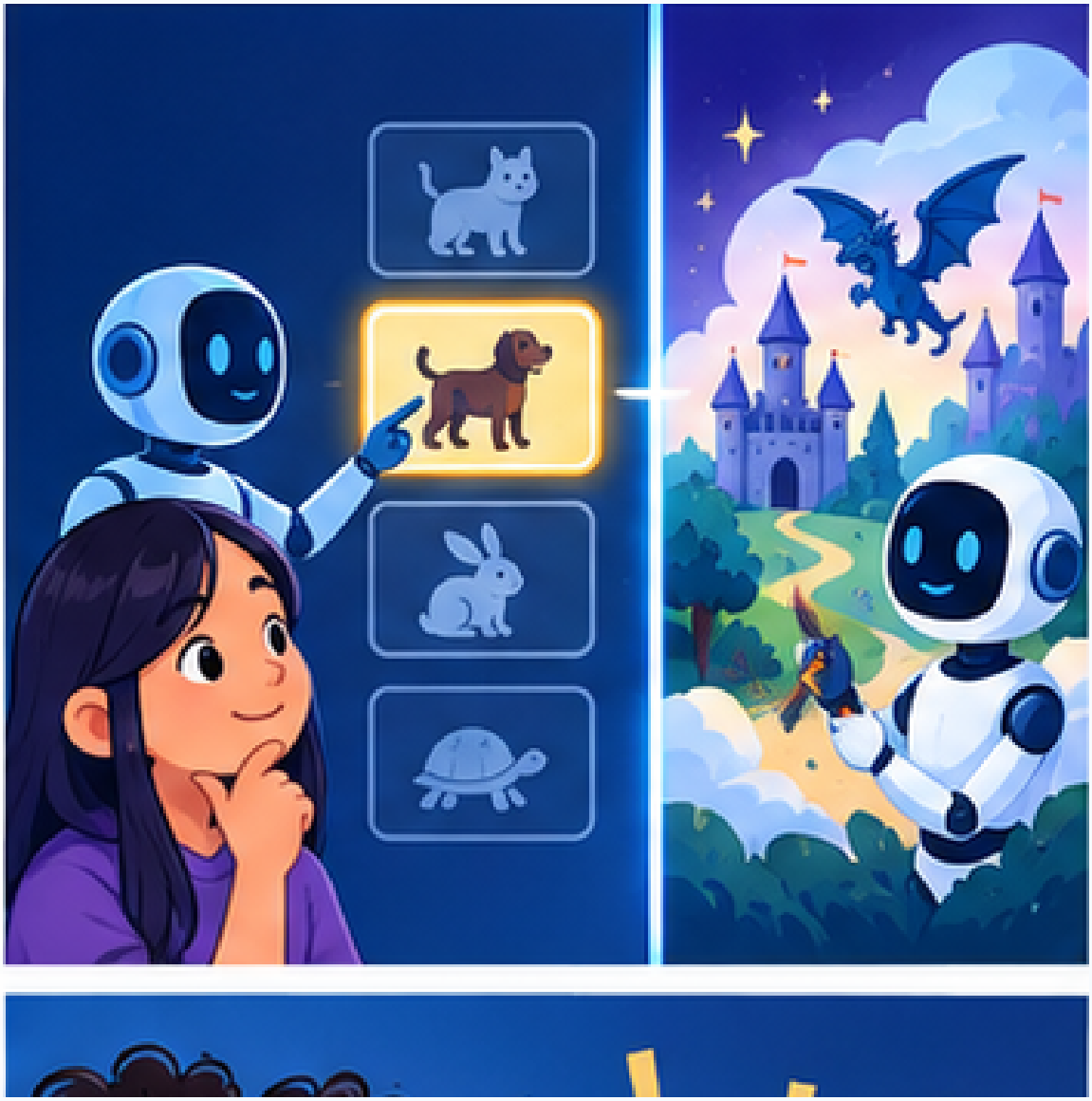
A fluent completion is still an **output** to evaluate against the task.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use predictive and **generative ai** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Compute what the supplied corpus says

> A conditional distribution is not an observation of the current marker.

Read the complete marker count table

Prefix	Continuation	Count
The marker is	red	6
The marker is	blue	3
The marker is	green	1

Compute probabilities and preserve their meaning

There are ten observed continuations. The empirical probabilities are red 6/10 = 0.6, blue 3/10 = 0.3, green 1/10 = 0.1. They are nonnegative and sum to 1.

A simple frequentist classroom model uses these relative counts as its next-token distribution. These probabilities describe the supplied corpus counts, not the current colour of any actual marker.

Trace corpus versus current evidence

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Ten corpus continuations	Count red 6, blue 3, green 1	Distribution 0.6, 0.3, 0.1
Prefix: The marker is	Select a next token	An output, not an observation
Current marker question	Check direct evidence	Colour claim needs support

The continuation model and a factual observation have different evidence sources.

Explain the deciding step

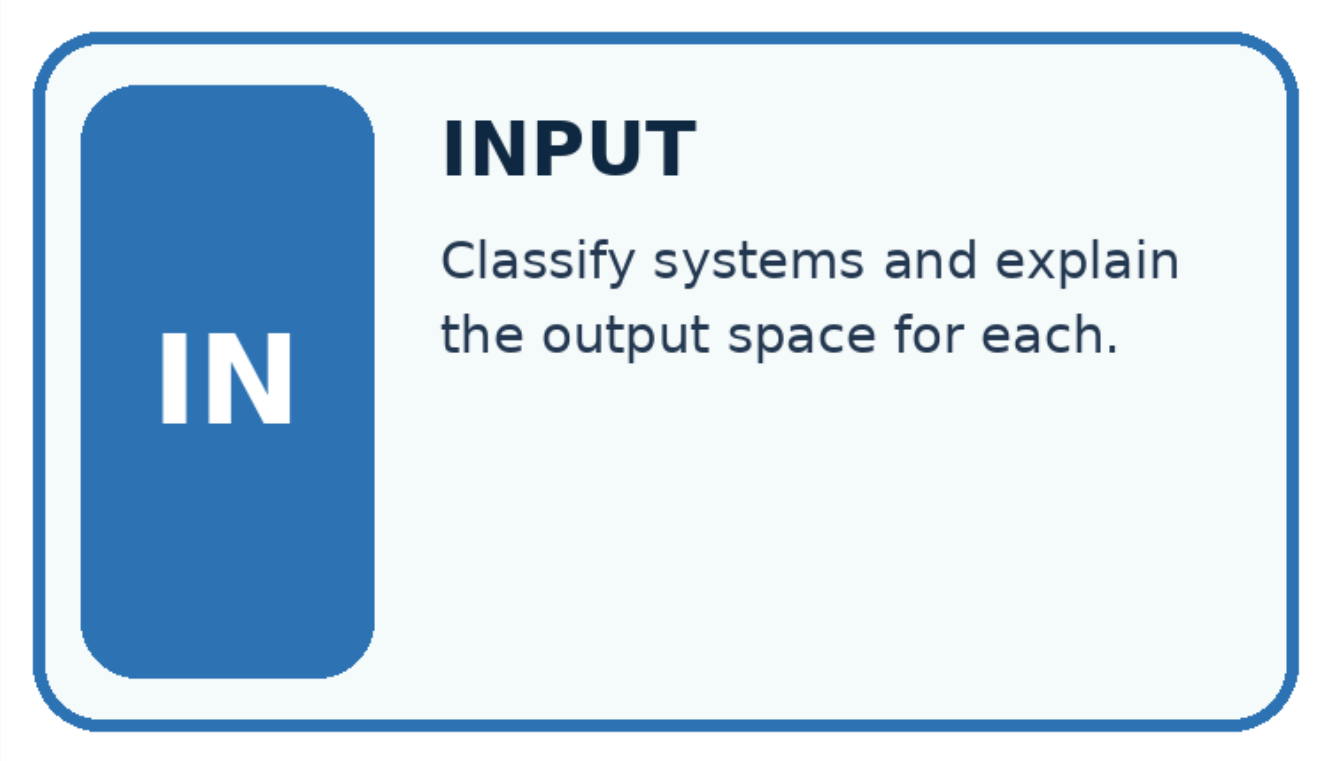
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

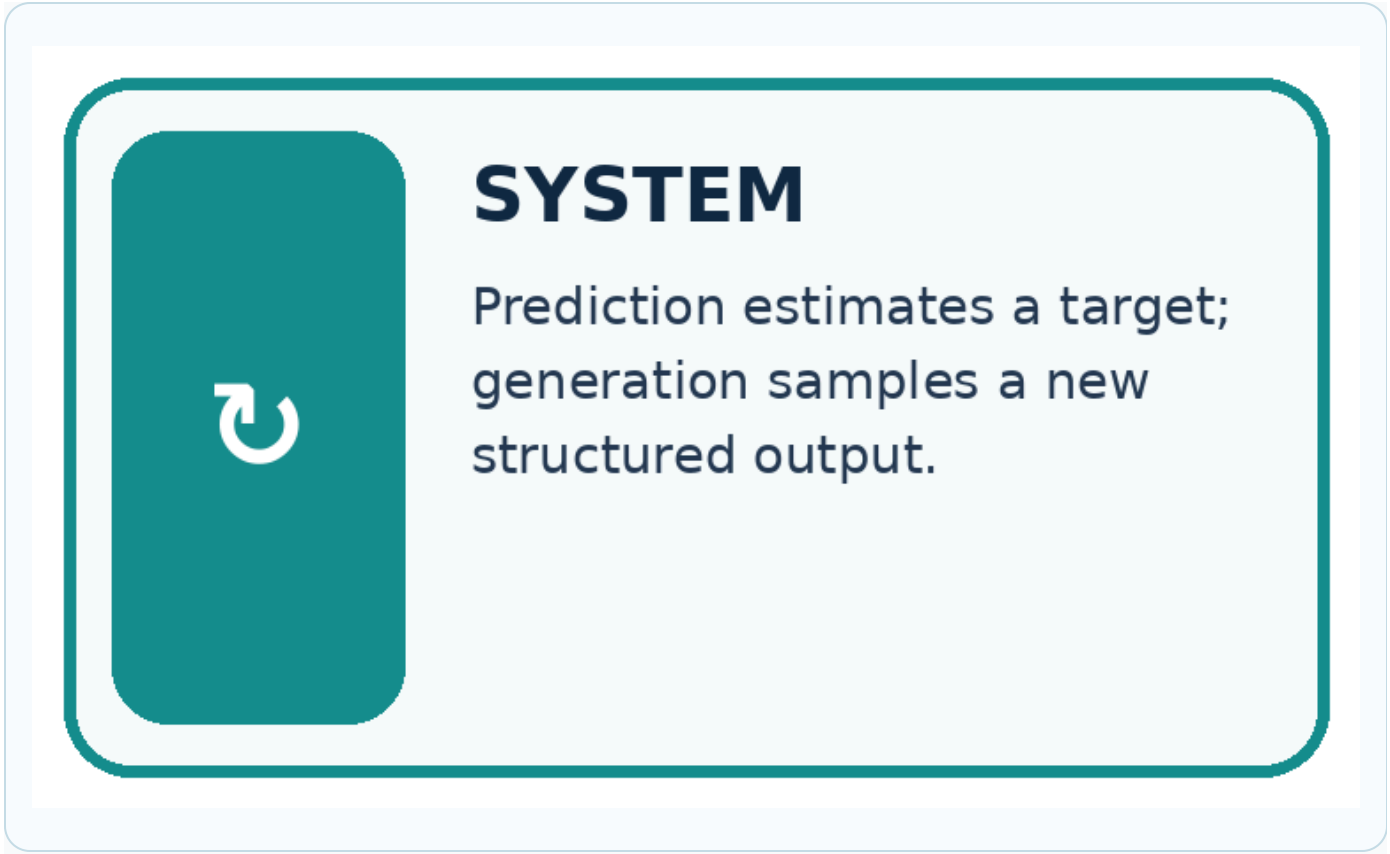
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Prediction estimates a target; generation samples a new structured output.
OUT
EVIDENCE
Classify systems and explain the output space for each.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

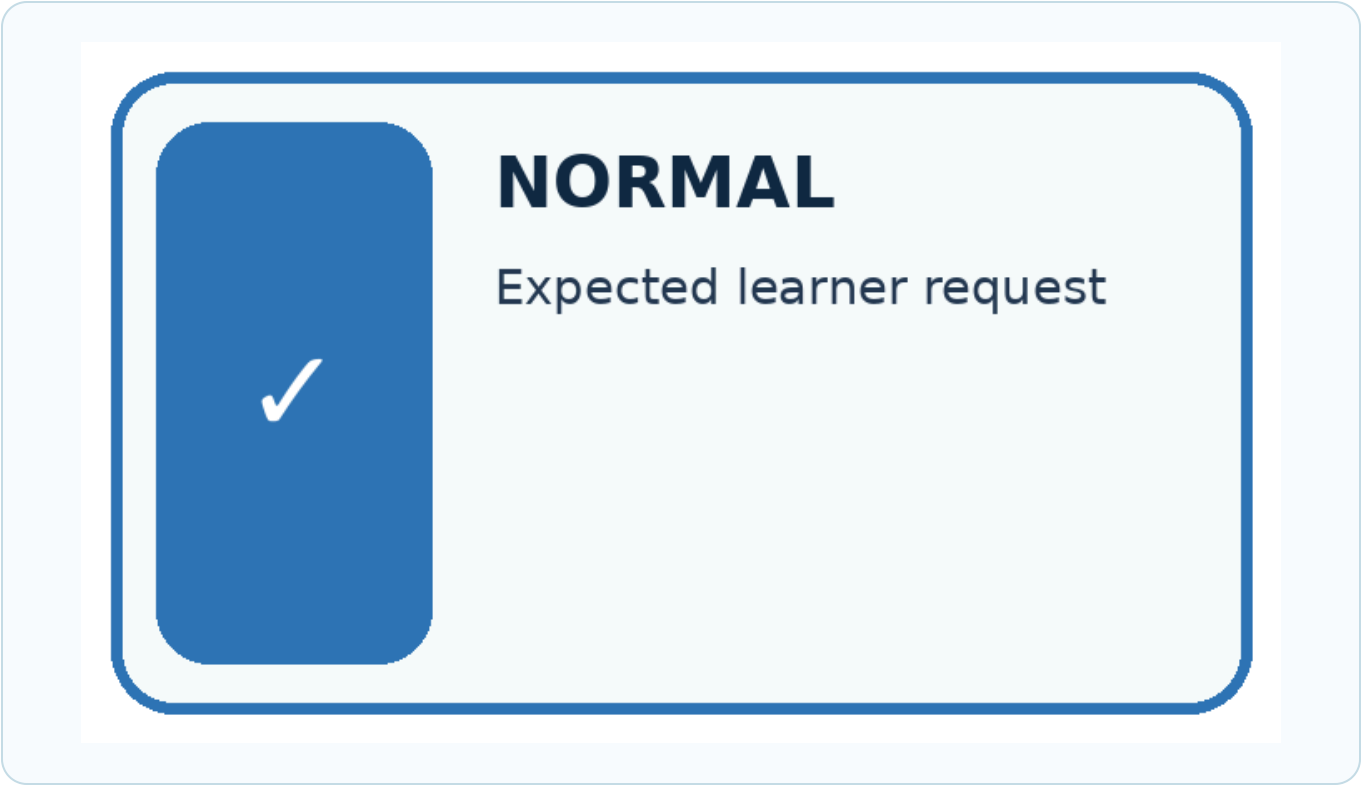
Connect the picture and the explanation

MAIN LESSON · SECTION 4

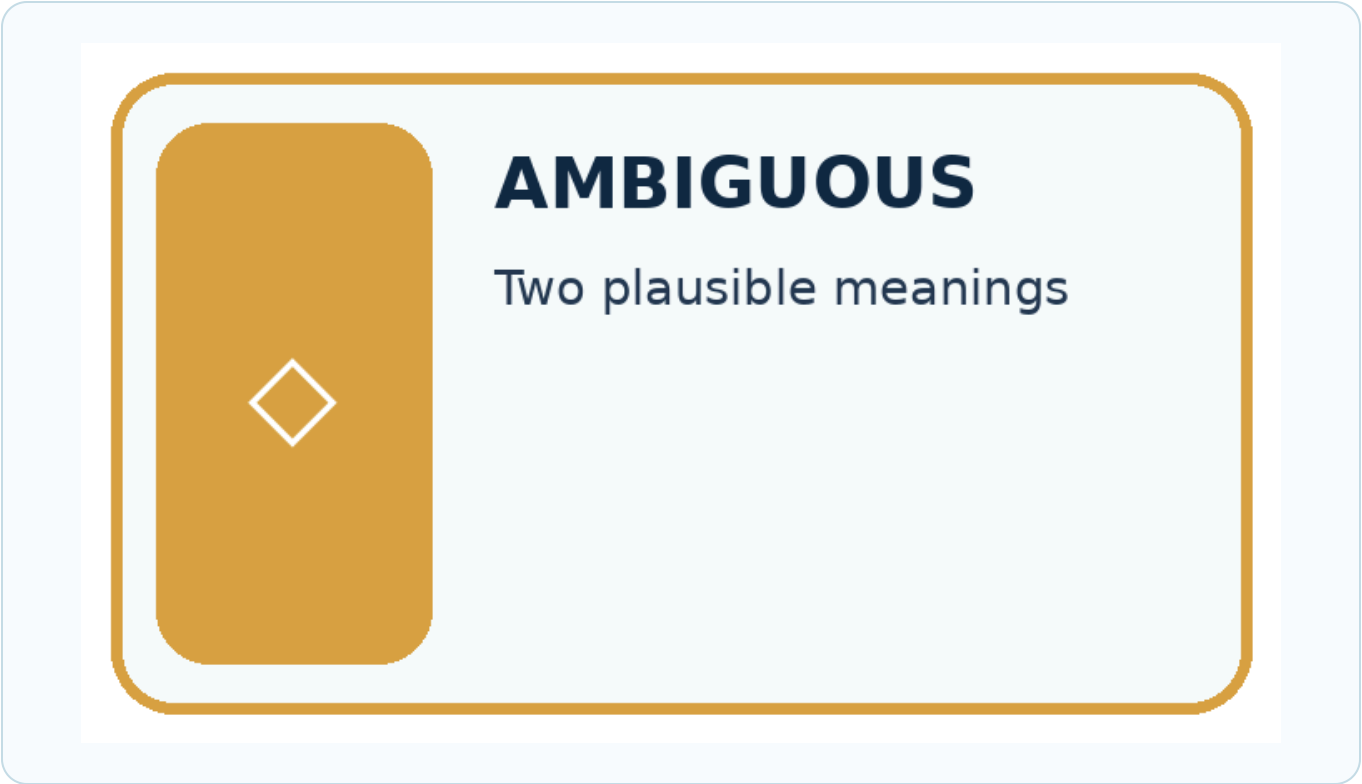
Compare mechanisms and evidence requirements

> A correct category does not validate a generated answer.

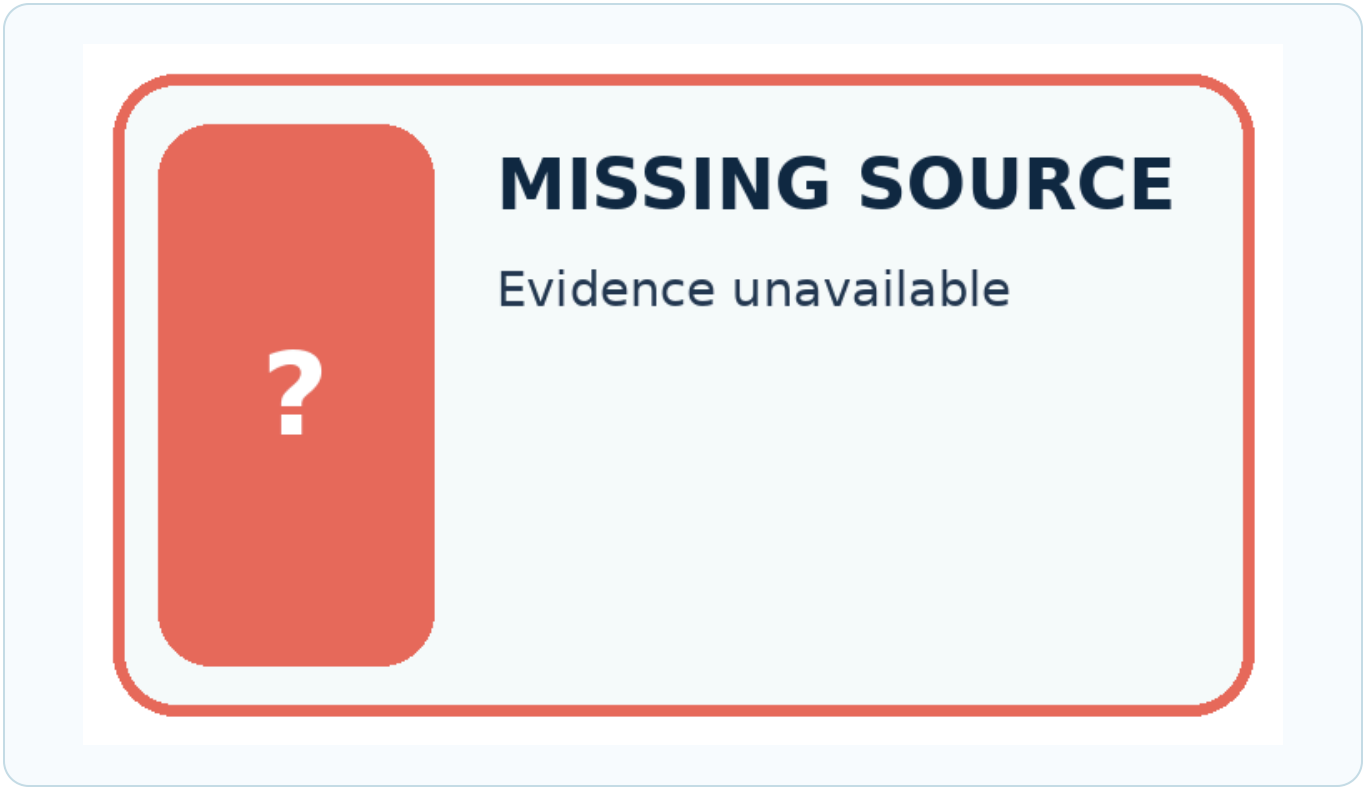
Four pictures with a purpose



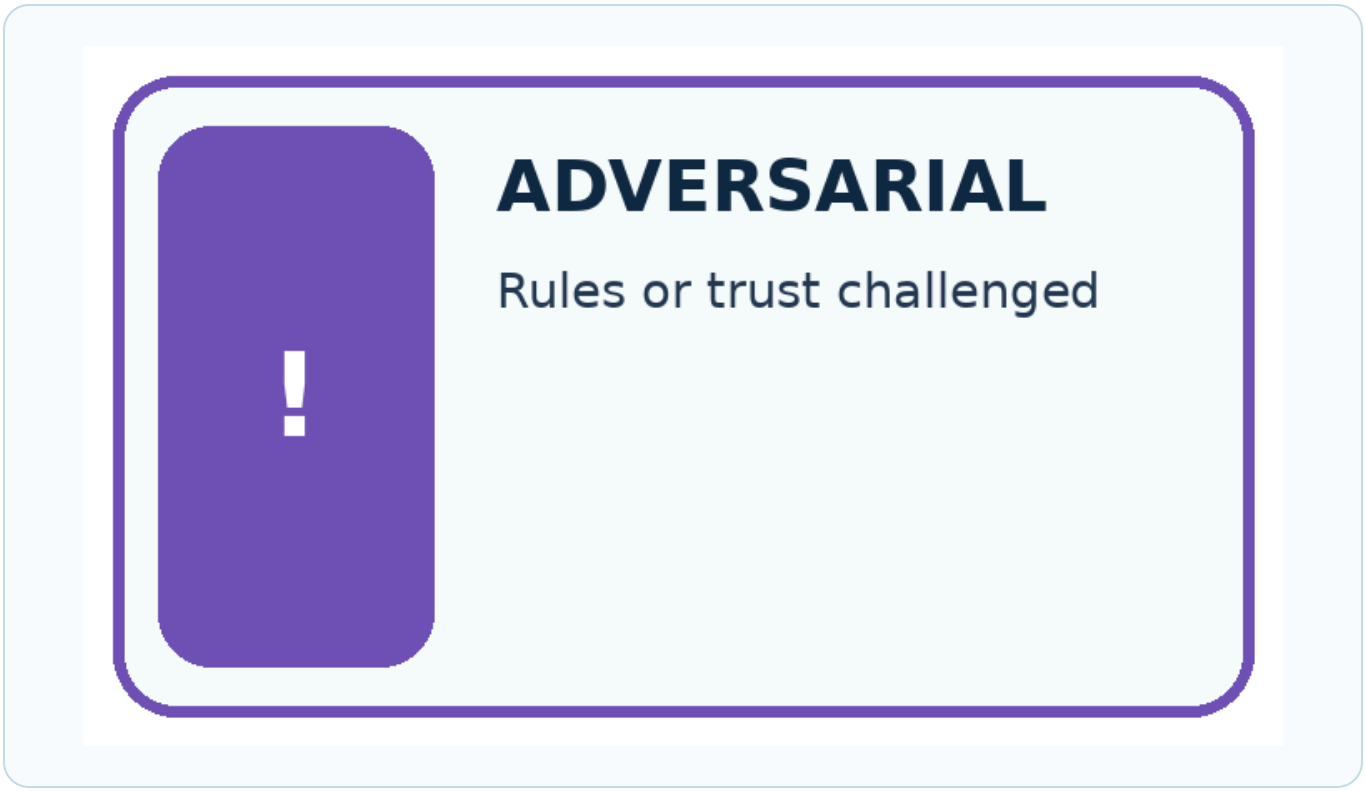
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: The classifier instead predicts history while N and the draft stay unchanged.

<details>
<summary>Check after predicting</summary>

Prediction: The category **prediction** changes, but the same unsupported draft claim still needs checking.

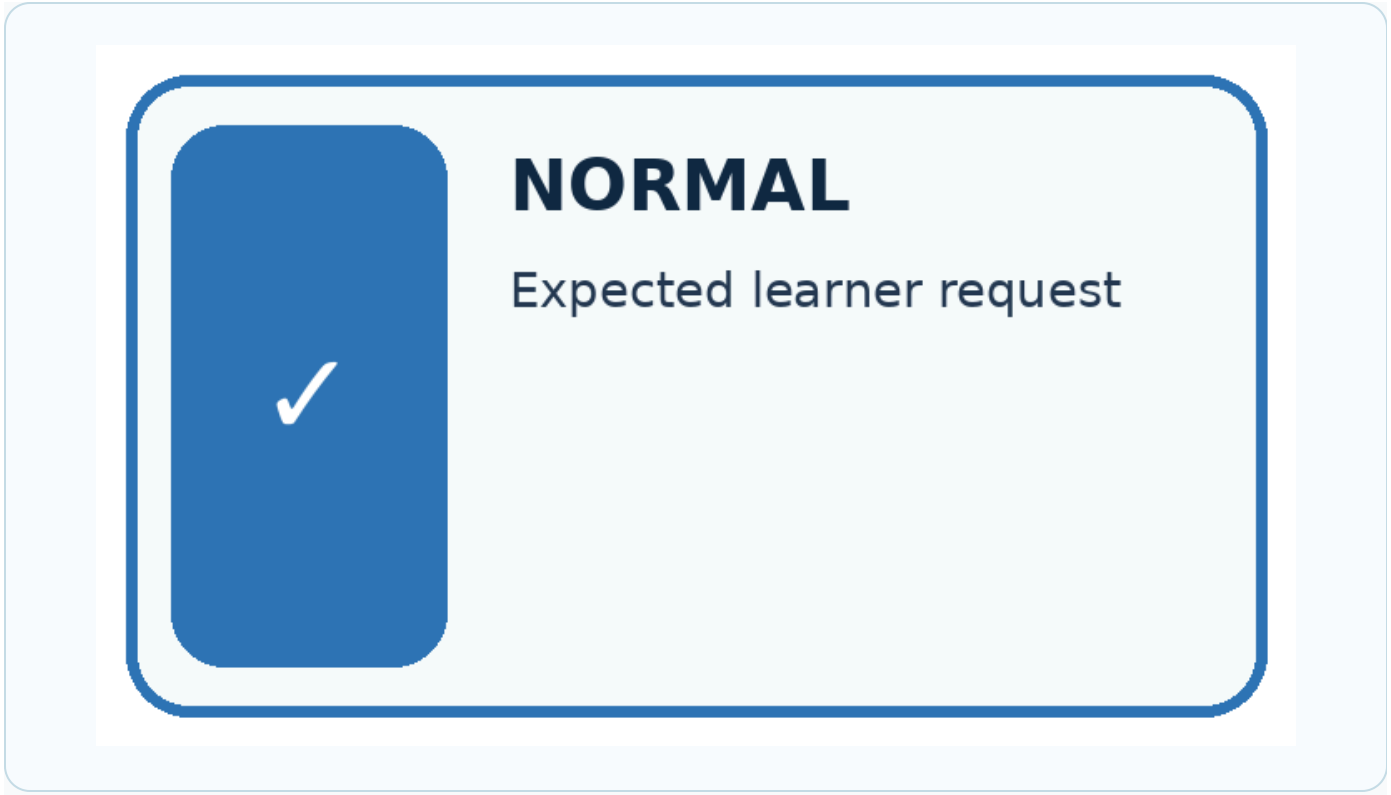
Reason: Stages can fail independently; one result does not validate another.
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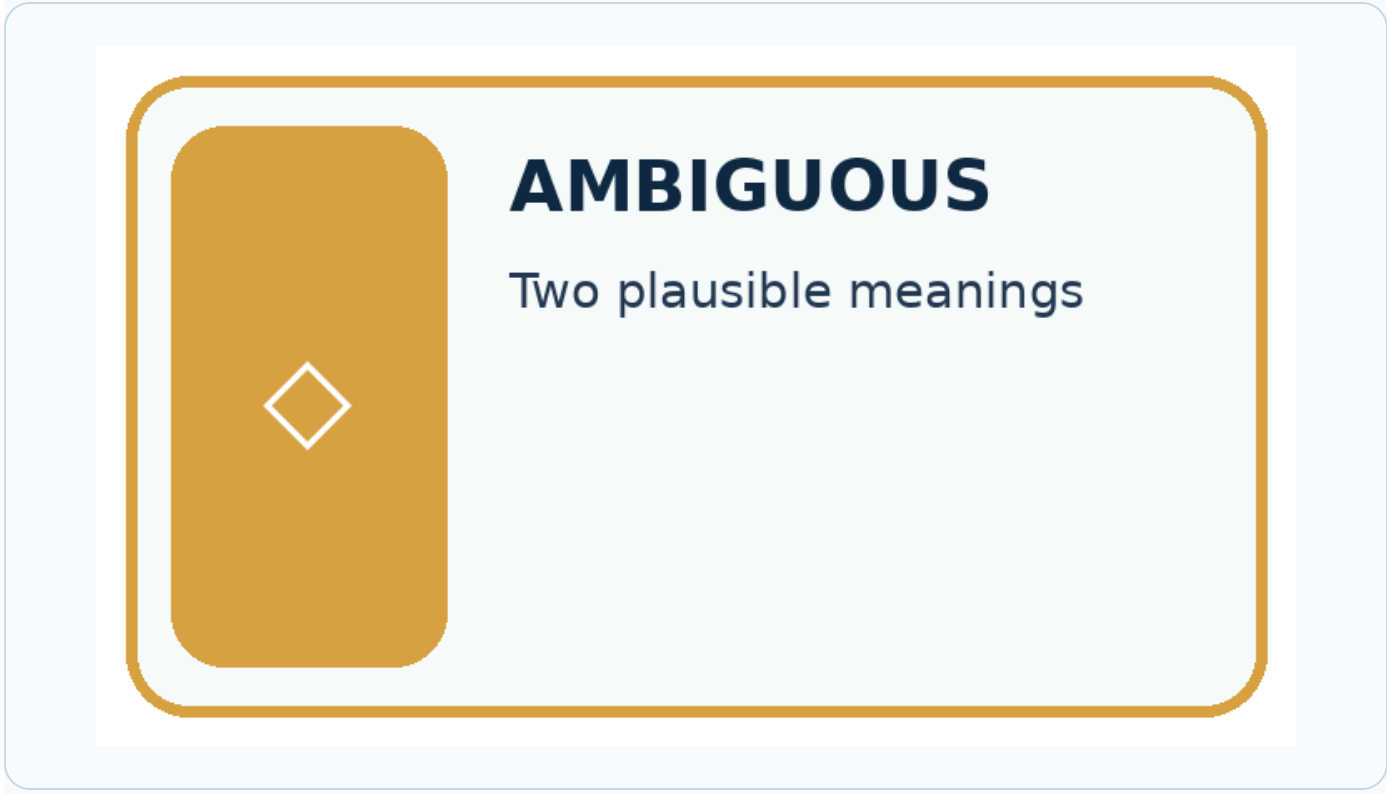
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<summary>Earlier wording, alternatives and picture transcripts</summary>

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.
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NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Expand the prefix, then audit a separate evaporation answer

> Do not confuse the next conditional table with the previous alternatives.

Trace the expanded-prefix distribution

After producing red, the next step conditions on the expanded prefix “The marker is red”. A second table might give an end token probability 0.8 and the token today probability 0.2.

A generated sequence can stop at the end token or continue. Do not reuse the first-step table unchanged after adding a token unless the model explicitly ignores the new prefix.

Separate text continuation from an image label

A separate classifier can return red as a label from an image; that is a different **input** and **target** from continuing text. A downstream evaluation asks whether the system correctly reports the current marker.

It must include ground truth or sources and appropriate failure handling, not just count plausible phrases.

Read the study-tool stages and approved note

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

A fitted classifier labels a question science; its **training** used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

Trace the unsupported addition

AI ACADEMY / AI-3/1 · 02 · TRACE THE CASE

Follow the deciding steps

01 Predict category

The classifier predicts science from the new question.

02 Retrieve note

N supplies a specific statement about evaporation.

03 Generate draft

The draft adds “always disappears forever” beyond N.

04 Verify claims

Keep the supported change-of-state claim; remove or review the unsupported addition.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Predict category	The classifier predicts science from the new question.
2. Retrieve note	N supplies a specific statement about evaporation.
3. Generate draft	The draft adds “always disappears forever” beyond N.
4. Verify claims	Keep the supported change-of-state claim; remove or review the unsupported addition.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

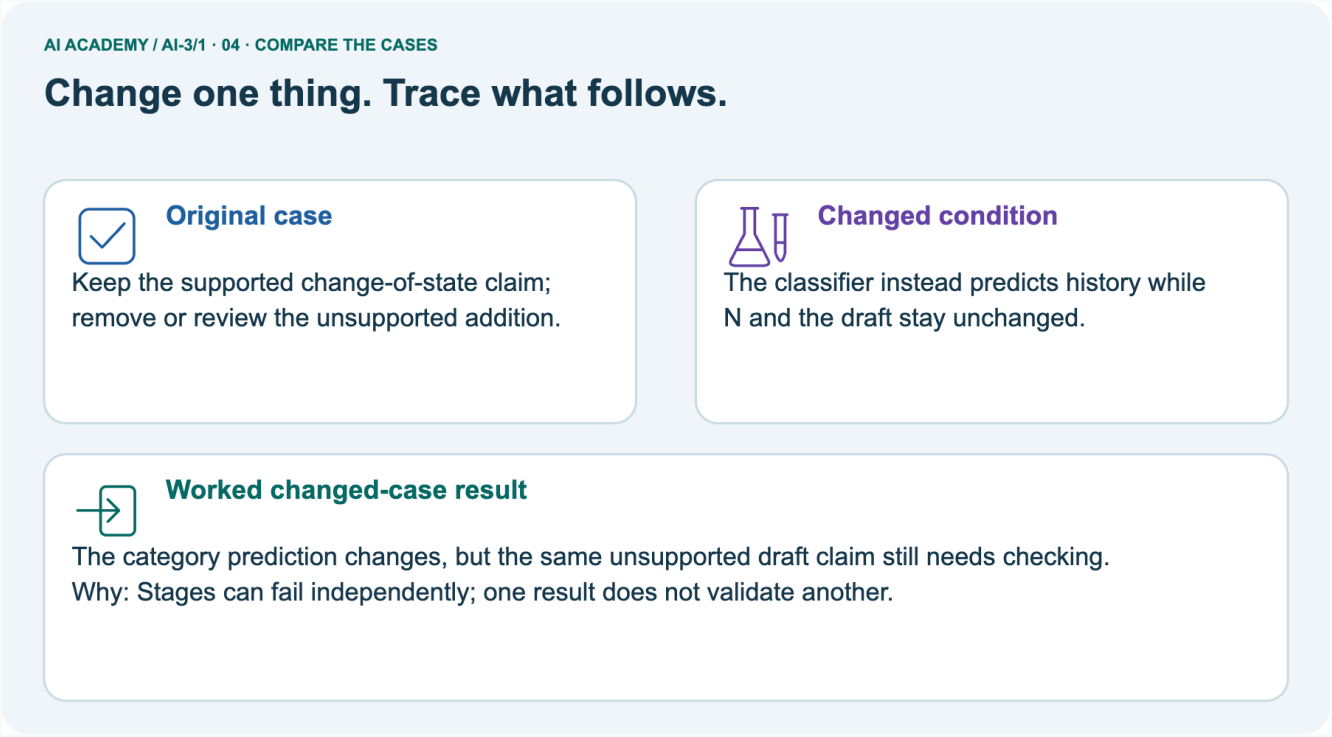
Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct.

Predict changing only the classifier label

- Change one thing: The classifier instead predicts history while N and the draft stay unchanged. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the evidence limit

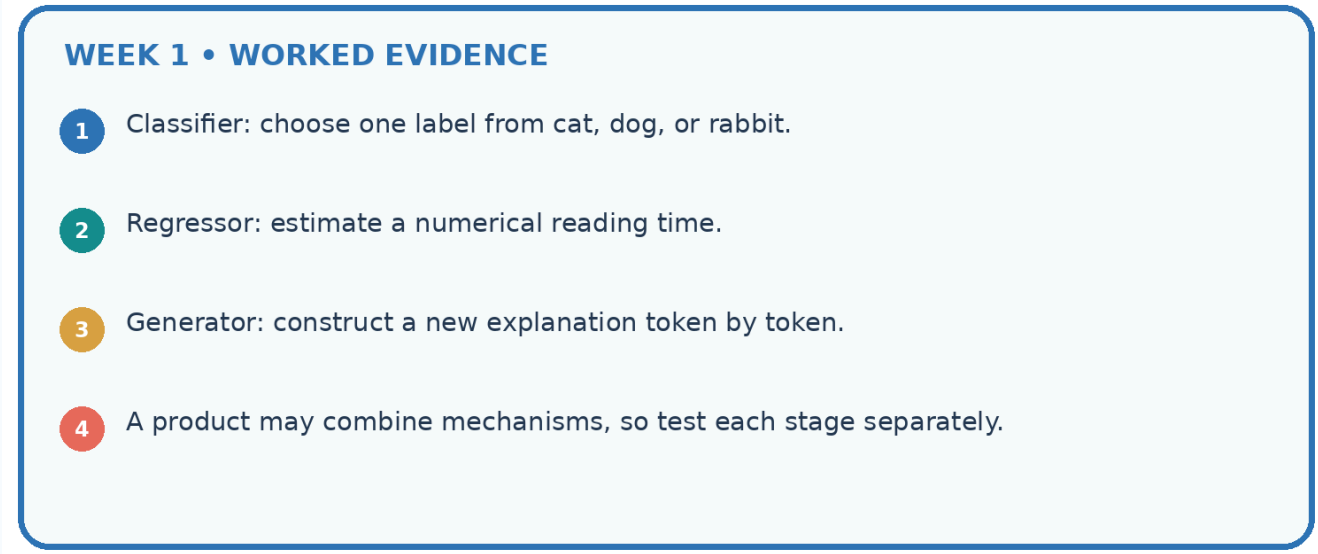
- This example does not expose a real model's architecture or prove that source-fed generation always stays grounded.
- Read the labelled picture: Identify the first unsupported claim and explain what source would be needed before asserting it.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Classifier: choose one label from cat, dog, or rabbit.

2. Regressor: estimate a numerical reading time.

3. Generator: construct a new explanation token by token.

4. A product may combine **mechanism**s, so test each stage separately.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

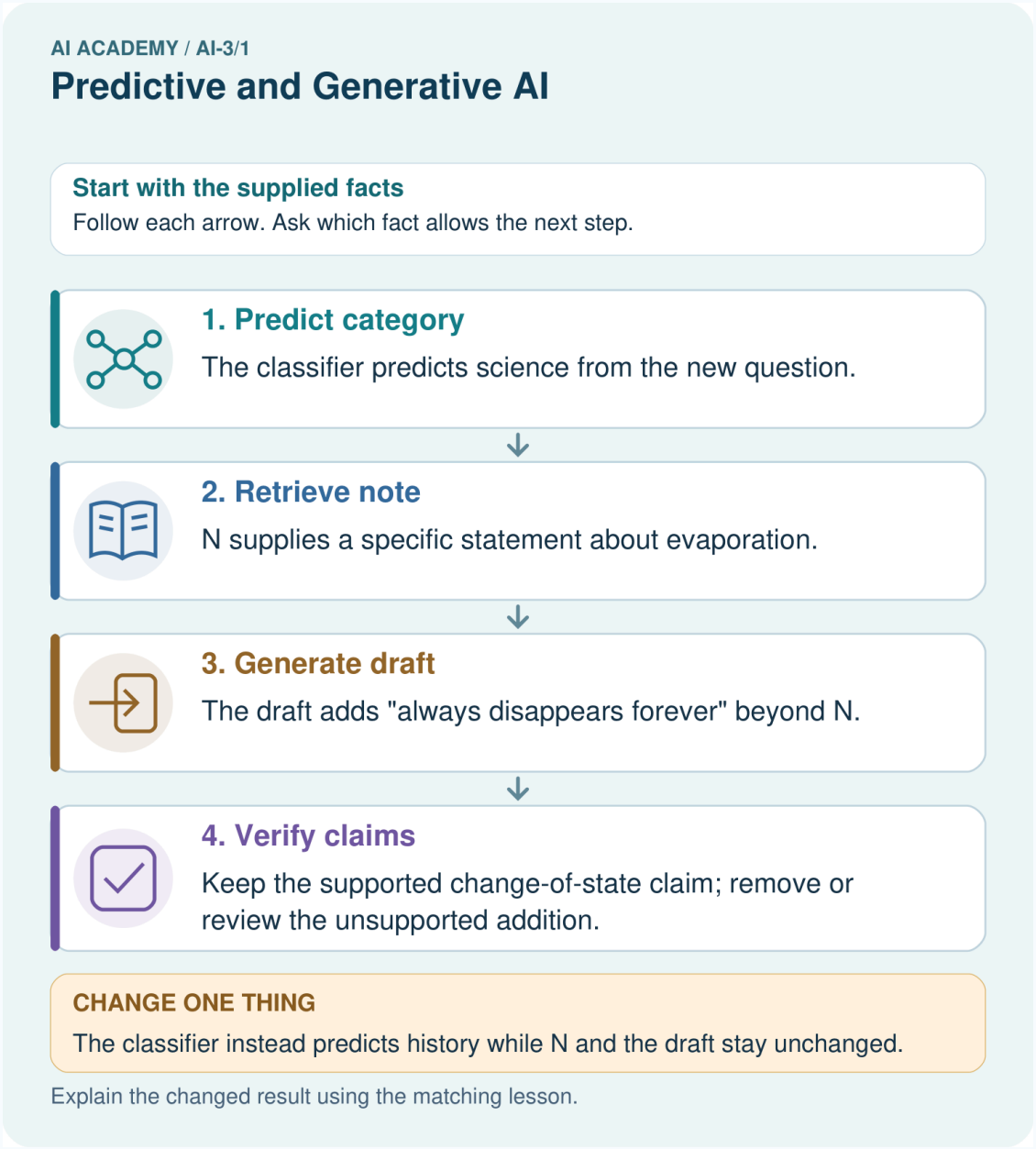
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Predictive and Generative AI

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Locate the first unsupported claim

> Check each stage using the evidence appropriate to that stage.

Repair the category-to-truth argument

Claim to repair: The question was classified science, so the whole generated answer is correct. The label is one output from one stage. Approved note N supports liquid water changing into water vapour; it does not support “always disappears forever”. Remove or review that addition rather than using a correct category or fluent wording as evidence.

A history label would change the category prediction without repairing the same unsupported draft.

Design a revealing stage-specific check

> **Exit: a team says “the highest-probability continuation proves the current fact”. Name the evidence they actually have and the evidence still needed.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Every AI system that produces an answer is generative. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Keep three questions separate: did the model fit continuation data; did the **output** satisfy the requested format or task; and are factual claims supported?

A system that drafts a fictional description has different criteria from one reporting a live measurement. Record the source and time when a claim refers to changing conditions.

No AI account is needed to reason about these distinctions.

MAIN LESSON · SECTION 8

Practise a separate door distribution with less help

> Keep corpus frequency and current measurement as different evidence.

Read the supplied door counts and observation

Supported case: the prefix “The door is” is followed in a tiny corpus by open four times and closed once. The empirical probabilities are 4/5 and 1/5. A current sensor reports closed.

Selecting the most frequent continuation open would be plausible under the corpus model but inconsistent with the sensor **observation**.

Calculate and explain the limit

> **Guided: compute both probabilities and explain why selecting open does not establish the current door state.**

Complete marker-table W2

Practise with less help: Compute the distribution

Use workbook W2 for the connected example “Generative and predictive objectives”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair generative models do not predict

- Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the separate evaporation picture

- Identify the first unsupported claim and explain what source would be needed before asserting it.

Explain the picture

> **My explanation using a pictured step and the given facts**

Keep independent work separate

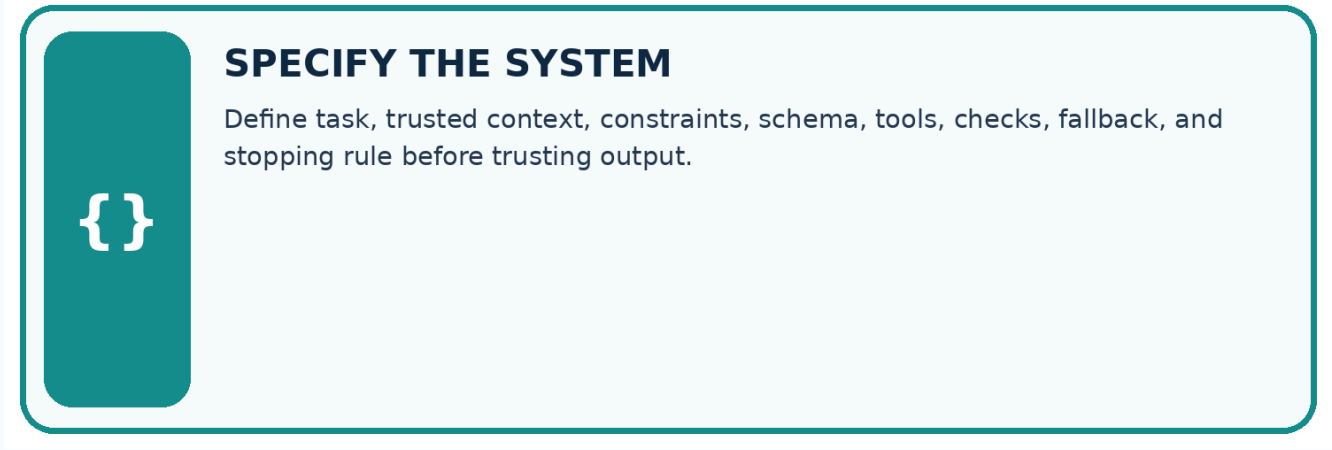
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Audit twelve complete component contracts

> Sort by task and evidence without assuming one category describes a whole app.

The twelve supplied cards

Card	Supplied task contract	Mechanism and evaluation cue
1	Written rule adds7and8	Calculation: expected56 from supplied integers
2	Lookup retrieves supplied note byID	Retrieval: exact ID, version and relevant passage
3	Learned message model outputs request/report	Classification: reviewed reference labels on separate cases
4	Learned model predicts numeric reading minutes	Numeric prediction: units and errors against separate references
5	Toy marker prefix model outputs a colour token	Generation: supplied conditional counts; not current marker truth
6	Generator drafts an evaporation explanation using N	Generation plus source check: every factual claim supported
7	Generator writes an acrostic from a supplied word	Generation: line initials and requested form; factual claims if present
8	Sequence transcriber outputs words from audio	Constrained sequence output: reference audio/transcript
9	Fixed rule ranks approved study activities	Rule-based ranking: eligibility, weights and tie policy
10	Classifier routes, lookup retrieves, generator drafts	Combined application: evaluate the stages separately
11	Search matches a query to a stored passage	Retrieval: relevance and source identity, not learned-generation proof
12	A person reviews a draft before publication	Human decision: review scope, content and intended destination

Identify targets, failures and checks

1. The twelve cards supply the missing investigation materials. For each identify input, output space, stated mechanism and a possible failure. The mechanism column is a supported cue: cover it for a first classification attempt, then compare your explanation. Do not infer a real architecture or performance score from the short task description.

Trace the source-grounded example and preserve scope

1. Trace the complete pictured note-and-draft example: keep the supported evaporation statement, flag the unsupported addition and explain why changing only the category label cannot fix it. Separately calculate the marker and door distributions from their own counts. These are supplied classroom contracts and paper traces; no model is trained or called.

   Reserve the Moon-note/difficulty/poem task for independent work before feedback.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing

> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → RUN → VERIFY → EXPLAIN

Sort twelve systems by mechanism. For each, name its output space, evidence needed, and one possible failure.

Use the twelve supplied component cards. Identify input/**output** contracts, stated **mechanism**s, possible failures and appropriate **evidence**. Cover the supported cue column for a first attempt. No real model **output** or performance measurement is supplied.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the Moon-note, difficulty and poem workflow

> Name each component’s target and the evidence its output would need.

Independent workflow and scope

A system retrieves an approved Moon note, uses a trained classifier to predict reading difficulty, then writes a poem. Identify each **mechanism** and the **evidence** needed to check its **output**.

Identify the three mechanisms and specify how you would check each. No actual Moon passage, classifier output, poem or reference difficulty labels are supplied for this planning question. Do not invent those results or claim measured correctness. The earlier signal-count exercise remains additional quantitative practice, not this selected assessment.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Start a study buddy whose stages can be checked

> Carry the prior level's evidence and permission boundaries into the new project.

This week's concrete contribution

For the Verified Study Buddy, draw routing/classification, source retrieval, calculation if needed, generation and claim checking as separate stages. Keep source IDs and exact passages alongside factual claims. A source supplied to a generator does not guarantee faithful output, and a correct route label does not establish the answer's truth.

Restrict the project to approved classroom material and preserve human review where required.

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record.

Keep the checkpoint and human choice

THIS WEEK'S CHECKPOINT Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

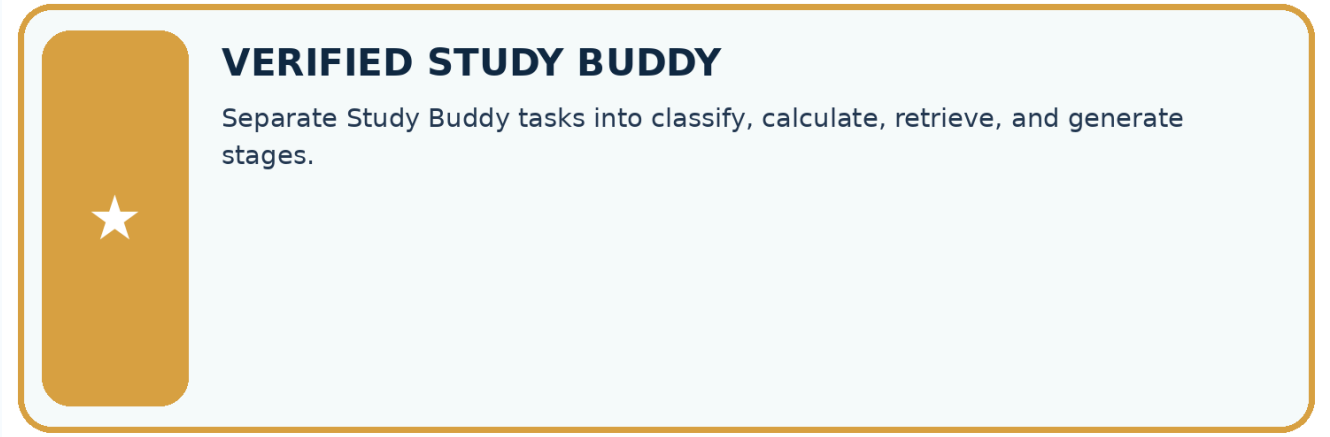
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: name the objective, output and evidence

> Plausibility and correctness need different checks.

Explain without notes

- Close your notes. Explain this goal in your own words: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
predictive AI	AI used to estimate a target such as a category, number or future outcome from inputs.	
generative AI	AI that produces new content such as text, images or audio rather than only selecting a category.	
target	The category, value or outcome a prediction task is intended to estimate.	
output space	The set or range of outputs a system can produce for a task.	

Terms for the connected explanation

Word	Meaning
Generative objective	A training aim concerned with producing or modelling possible outputs
Discriminative objective	A training aim mapping inputs to a specified target such as a label

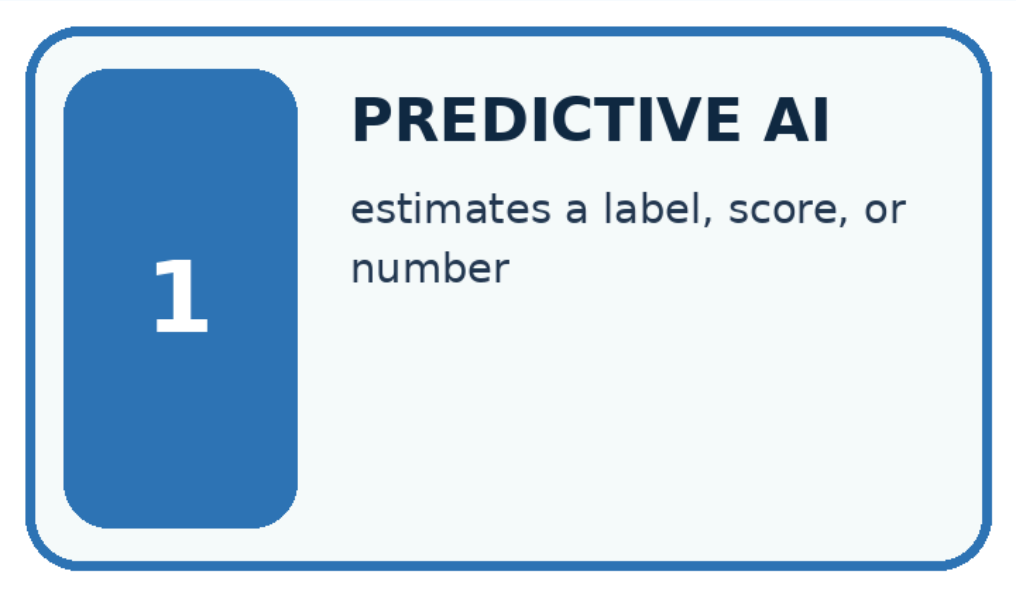
Word	Meaning
Conditional distribution	Probabilities for alternatives given stated information
Autoregressive generation	Producing a sequence one next item at a time using preceding items

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

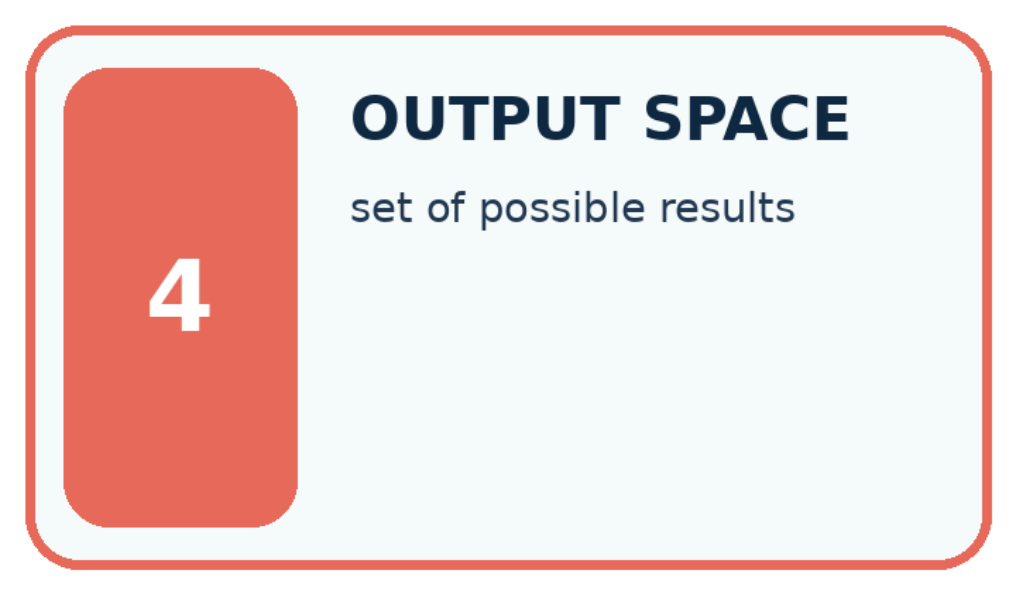
GENERATIVE AI

constructs new content
from learned patterns

3

TARGET

value a predictive system
estimates



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

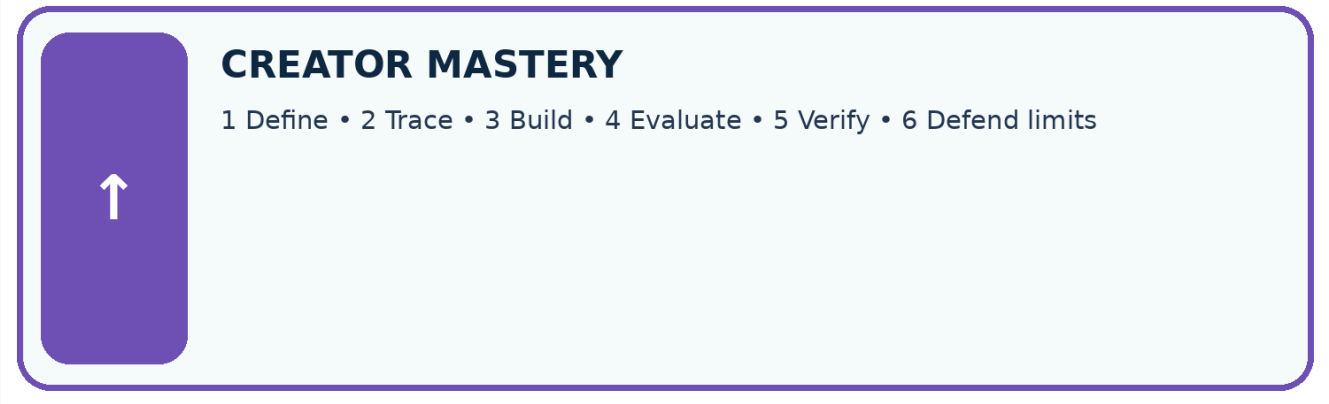
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit a new study workflow independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

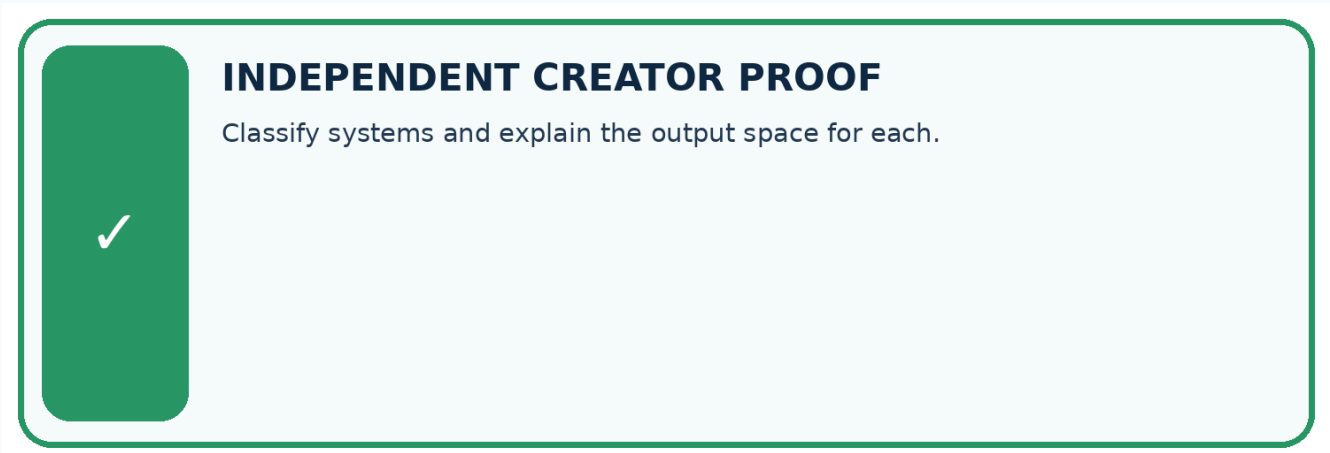
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit a new study workflow independently is the selected independent case.



EXPLANATION CHECK Identify retrieval, fitted category prediction and generation, and specify evidence without inventing unavailable **output**s.

Write or draw your reasoning and evidence.

▼ **Extra practice: Three outputs need three different checks**

REFERENCE · SECTION 13

Extra practice: Three outputs need three different checks

> Try the question before opening its worked explanation.

Read the supplied case and question

A tool labels a question biology, writes a new question and computes 7 times 8.

Identify each operation.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

Evidence or stage	Value or result
Classify	Select biology as a category
Generate	Compose a new question
Calculate	7 times 8 equals 56

▼ Earlier wording, alternatives and picture transcripts

Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Three outputs need three different checks

A tool labels a question biology, writes a new question and computes 7 times 8.

Identify each operation.

Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Distinguish discriminative prediction from generation by inspecting the requested output	Explain the deciding step, show a trace or test, and state one limit.
Calculate an empirical next-token distribution from counted continuations	Explain the deciding step, show a trace or test, and state one limit.
Separate next-token fit from factual support and downstream usefulness	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional learning tool first classifies a question, then drafts an answer from a supplied note.
- Which labelled step matches this detail? Keep the supported change-of-state claim; remove or review the unsupported addition.
- 1. Predict category
- 2. Verify claims
- 3. Retrieve note

- 4. Generate draft

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the first explanation

After your attempt, compare with the supported explanation: Verify claims Keep the supported change-of-state claim; remove or review the unsupported addition. Classification predicts a category under a trained mapping.

Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct.

Changed-case question

- Changed-case practice: The classifier instead predicts history while N and the draft stay unchanged. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. The category **prediction** changes, but the same unsupported draft claim still needs checking.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The category **prediction** changes, but the same unsupported draft claim still needs checking. Stages can fail independently; one result does not validate another.

Repair and save your work

Review Capstone Expo (ai-2/36). Goal: Defend an application using reproducible **evidence**. Short practice: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

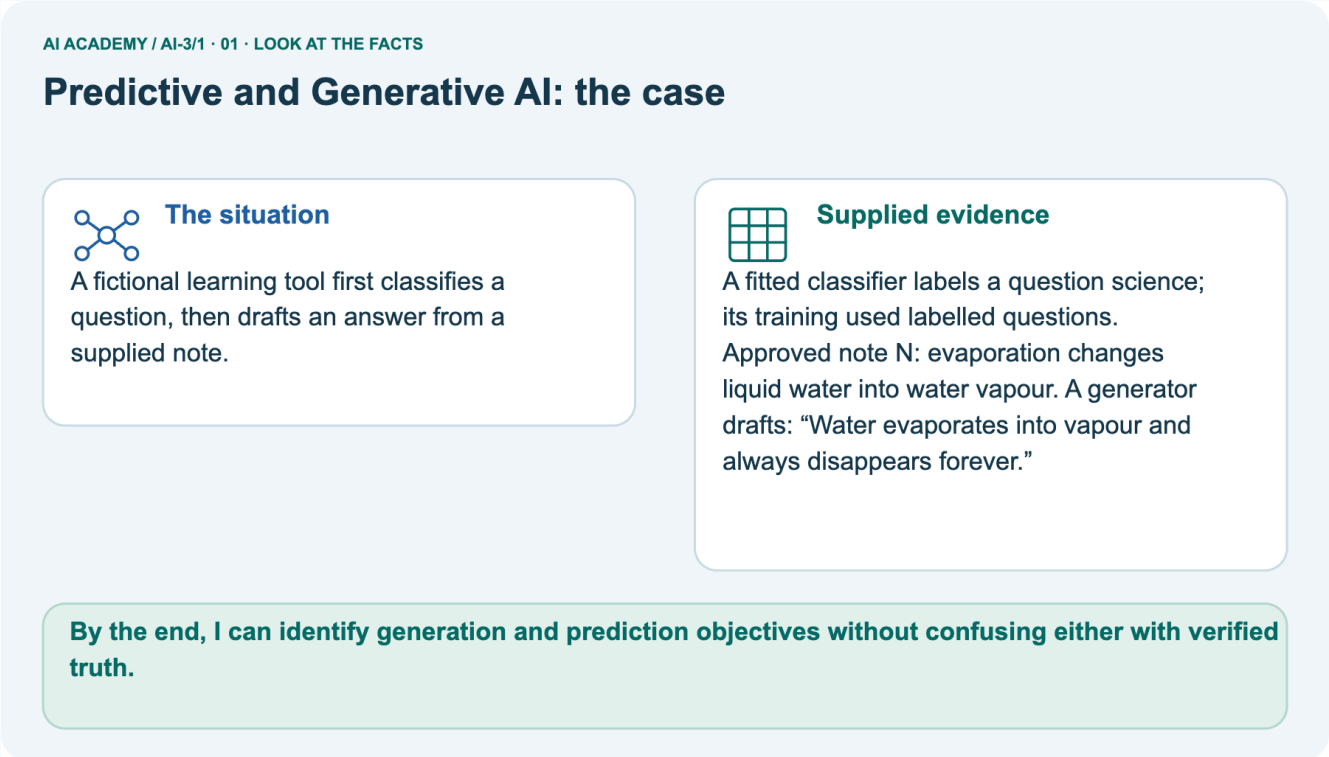
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Predictive and Generative AI: the case



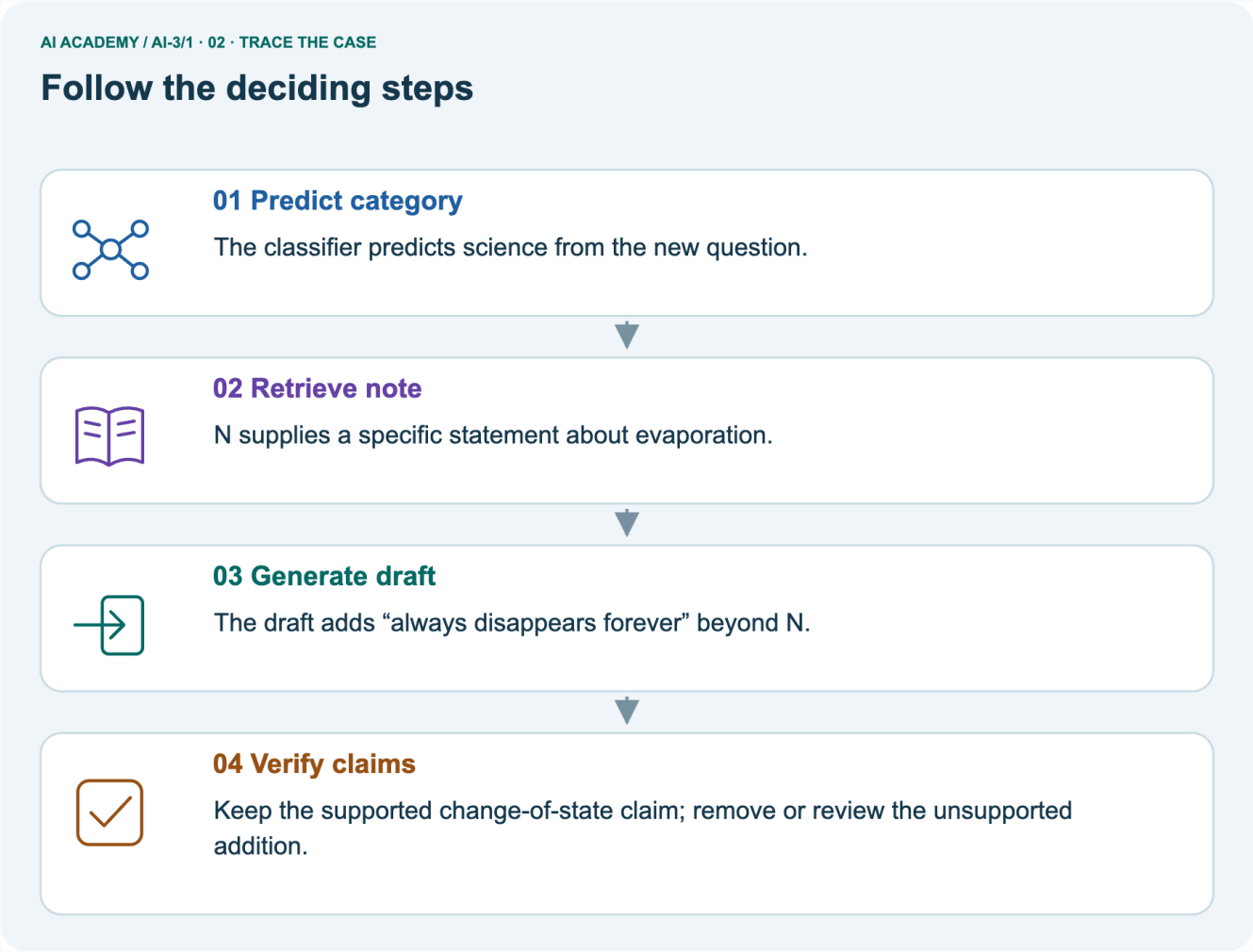
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional learning tool first classifies a question, then drafts an answer from a supplied note. Supplied facts: A fitted classifier labels a question science; its **training** used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

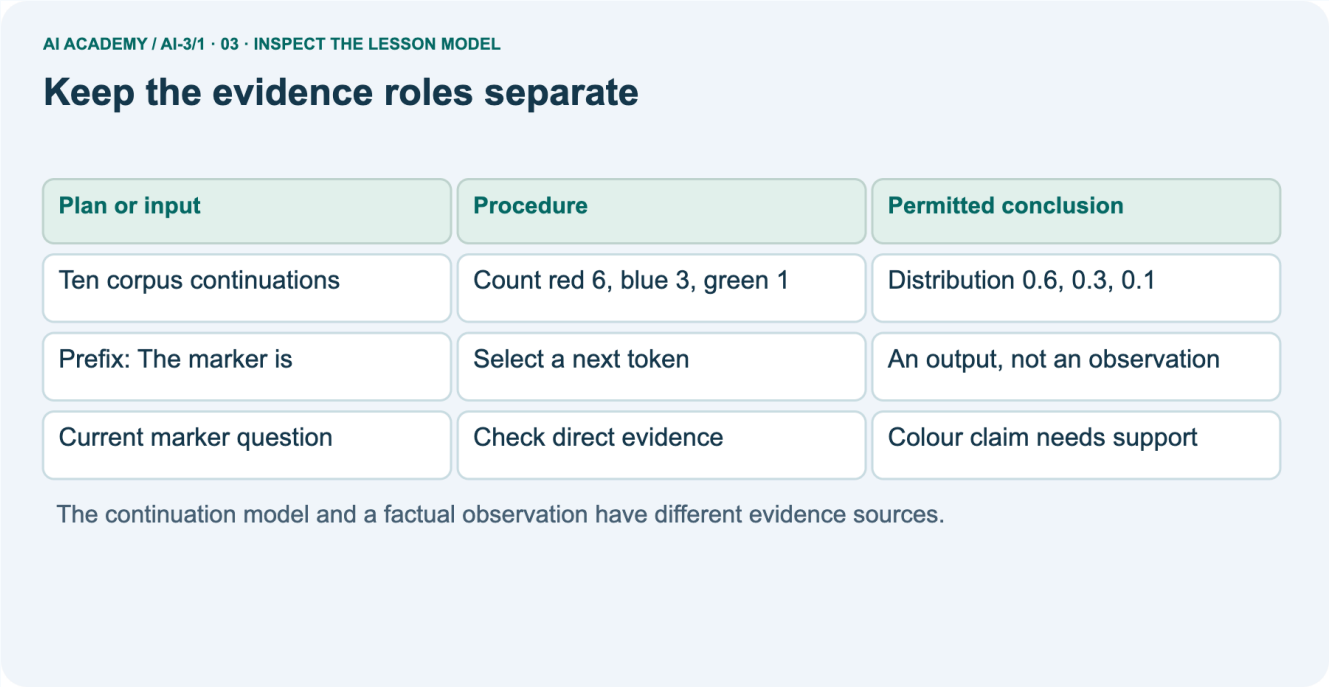


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict category: The classifier predicts science from the new question. → Retrieve note: N supplies a specific statement about evaporation. → Generate draft: The draft adds “always disappears forever” beyond N. → Verify claims: Keep the supported change-of-state claim; remove or review the unsupported addition.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep the evidence roles separate



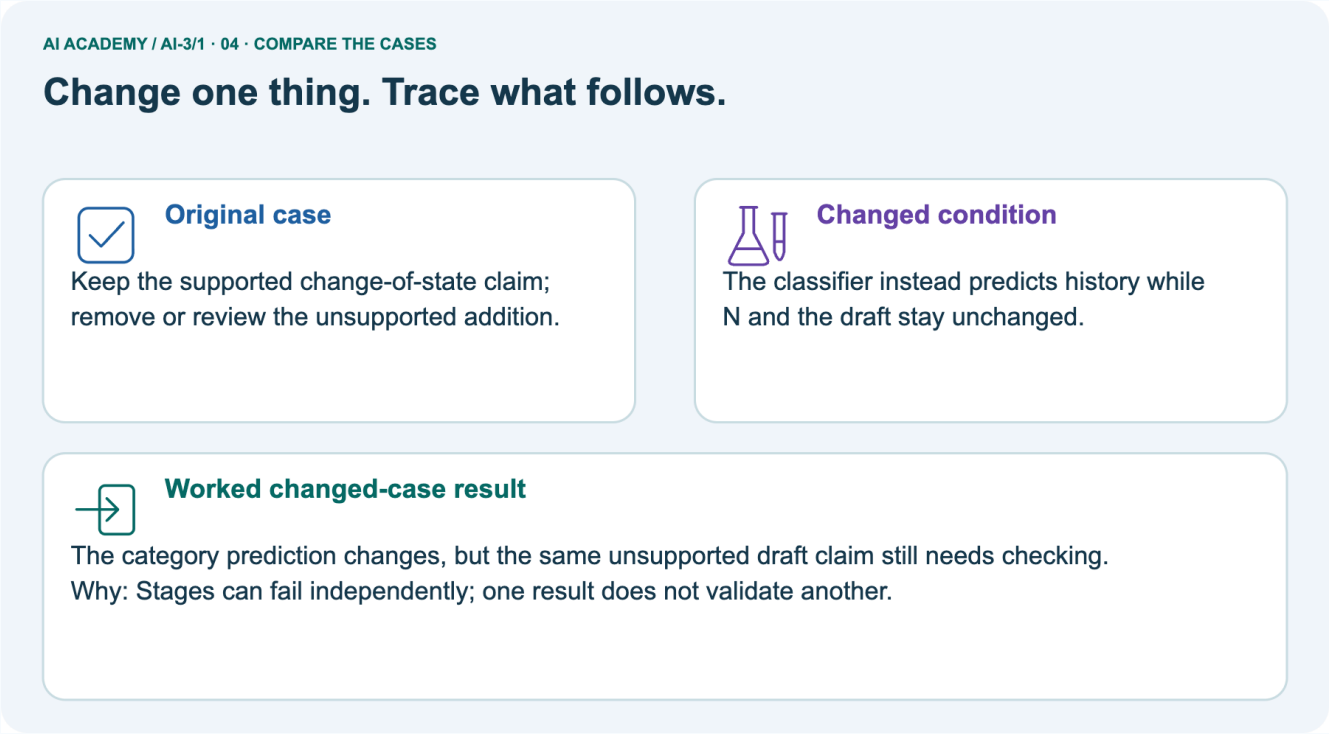
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. Ten corpus continuations | Count red 6, blue 3, green 1 | Distribution 0.6, 0.3, 0.1. Prefix: The marker is | Select a next token | An **output**, not an **observation**.

Current marker question | Check direct **evidence** | Colour claim needs support

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



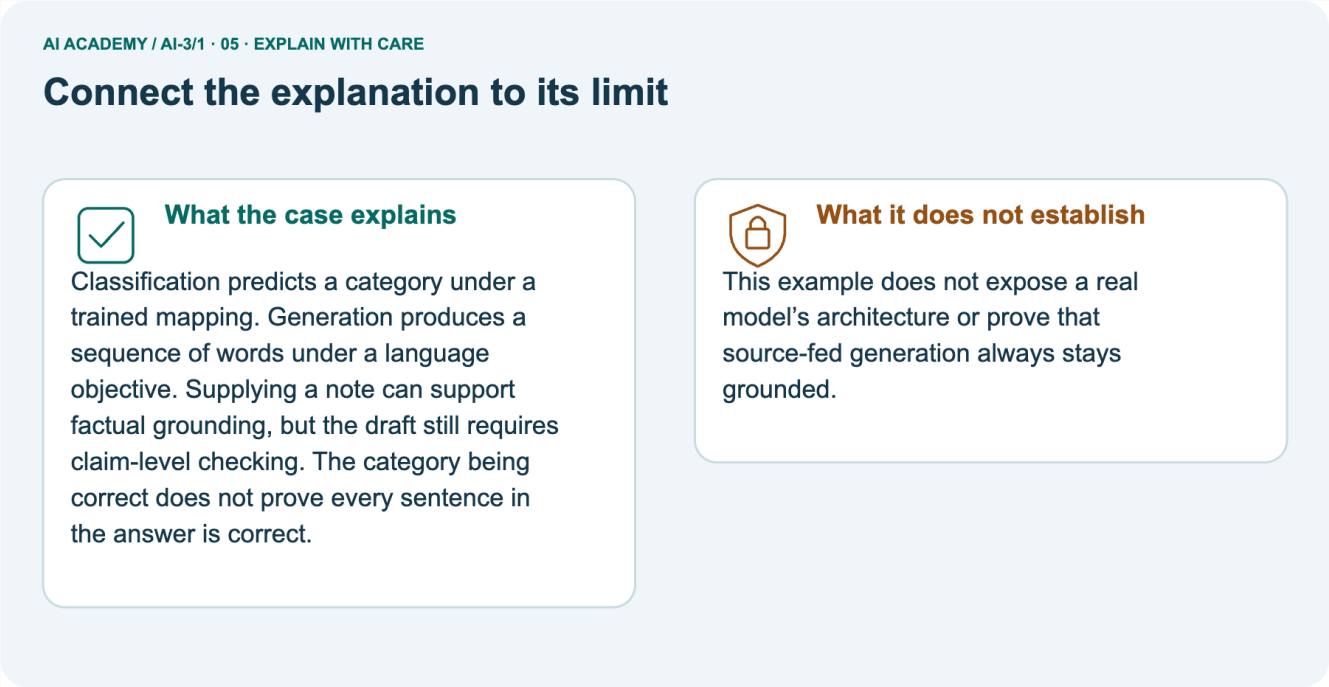
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep the supported change-of-state claim; remove or review the unsupported addition. Changed condition: The classifier instead predicts history while N and the draft stay unchanged.

Expected result: The category **prediction** changes, but the same unsupported draft claim still needs checking. Reason: Stages can fail independently; one result does not validate another.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



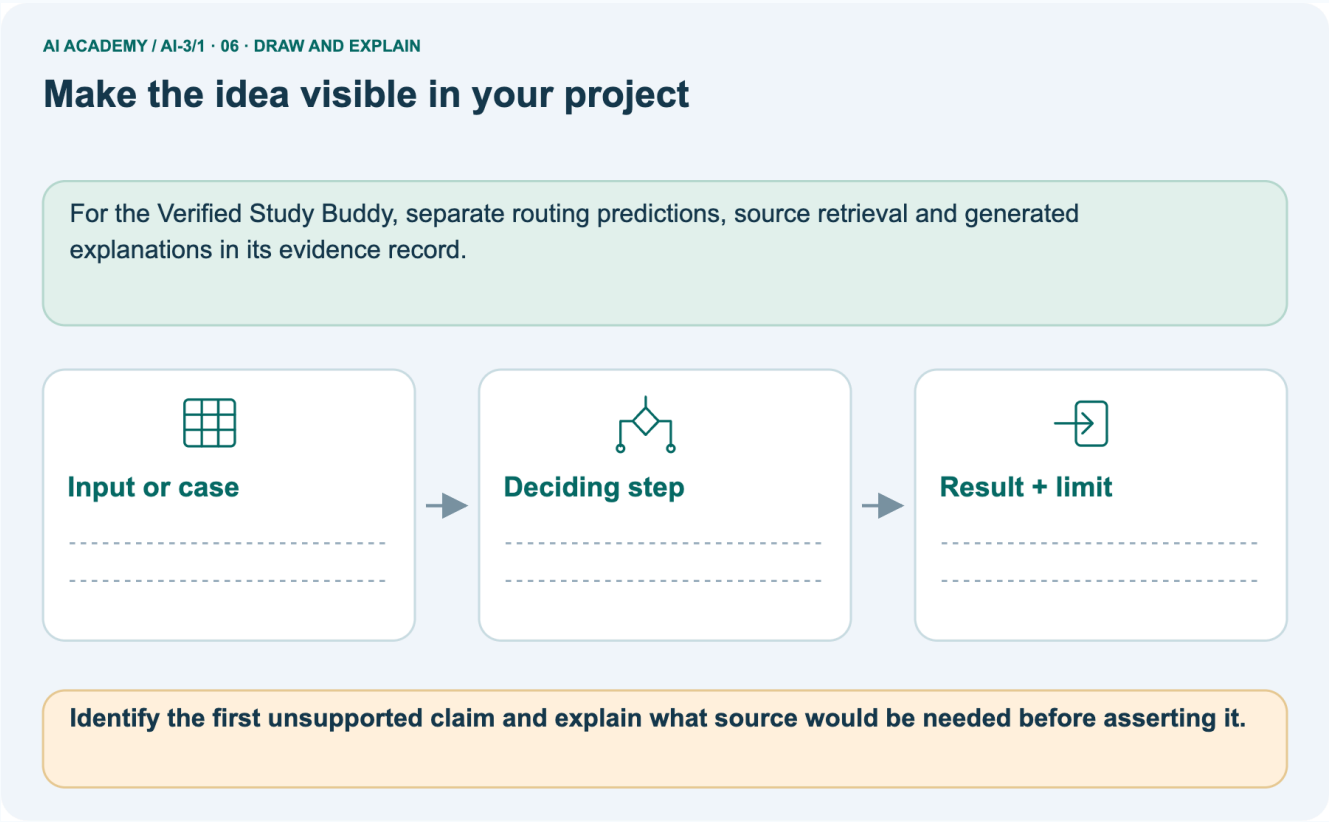
Follow the labelled picture with the matching explanation. Open the vector diagram

Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct. Limit: This example does not expose a real model’s architecture or prove that source-fed generation always stays grounded.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record. Draw your **input**, deciding step, result and limit.

Supported question: Identify the first unsupported claim and explain what source would be needed before asserting it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Predictive and Generative AI: the case

02. Follow the deciding steps

03. Keep the evidence roles separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.